

Supplementary Tables

All twelve tables produced by the analysis pipeline. Every table is regenerated by code/reproduce_all.py and is also available as CSV and Markdown in tables/. Units are dimensionless unless stated; energies are in squared signal units; ε and η are ratios; SNR is in dB; abbreviations follow the manuscript (η retained energy, ε_{L2} relative L2 error, ε_P Parseval discrepancy, AC mean-removed basis).

Table S1. Recovered dataset summary and recovery validation evidence.

Source: tables/table01_recovered_dataset_summary.csv

field	value
n_samples	26
max_imag_part	9.841016890277388e-13
residual_vs_printed_m in	-0.06346982522626377
residual_vs_printed_m ax	-0.06229787027868383
residual_spread_ptp	0.001171954947579934
residual_mean_offset	-0.06273943395144102
dc_rounding_predicte d_offset	-0.06274965105625545
max_abs_residual_aft er_rounding	0.0
max_distance_to_inte ger	0.06346982522626377
printed_k14_equals_c onj_k12	True
printed_nyquist_imag	2.154e-12
dc_printed	1256.0

field	value
dc_implied	1256.319961695207
sequence	[251, 214, 195, 175, 174, 173, 198, 295, 403, 388, 318, 288, 232, 203, 215, 303, 212, 228, 135, 165, 227, 286, 316, 296, 216, 300]
mean	246.3846153846154
std_population	66.63914126044088
r_mean_over_std	3.6972957742910944
minimum	135.0
maximum	403.0
sum	6406

Table S2. Independent verification of the 2013/2014 results.

Source: tables/table02_historical_verification.csv

quantity	historical	recomputed	units	rel_diff	status	note
E (sample domain)	1.694e+06	1.6938e+06	-	0.000118064	verified	NaN
H (transform domain)	1.694e+06	1.6938e+06	-	0.000118064	verified	NaN
E1 (k=5, 11 coefficients)	1.68e+06	1.67982e+06	-	0.000106414	verified	NaN
100*E1/E	99.175	99.1747	%	2.93187e-06	verified	NaN
E2 (all coefficients)	1.694e+06	1.6938e+06	-	0.000118064	verified	NaN
100*E2/E	100	100	%	1.42109e-16	verified	NaN
0.95E	1.609e+06	1.60911e+06	-	6.83654e-05	verified	NaN
yc[k], ys[k] for k =	0	0.319962	-	0.319962	verified	yc[0] deviates by

quantity	historical	recomputed	units	rel_diff	status	note
0..14 (max abs deviation)						0.320 because it is printed to 4 s.f.
printed samples x[0..14] (max abs deviation)	0	0	-		0 verified	NaN

Table S3. Signal corpus definitions and generating parameters.

Source: tables/table03_signal_corpus.csv

id	name	provenance	N	mean	std	r	description	params
S0	Recovered historical baseline	recovered	26	246.385	66.6391	3.6973	The 26-sample sequence of the 2013/2014 study, recovered by Hermitian inversion of its published coefficients. Physical identity, units and sampling rate unknown.	{"N": 26, "source": "Udeh (2013/2014)", "recovered": true}
S1	Bin-centered sinusoid	synthetic	26	-8.54018e-18	0.707107	-1.20776e-17	Sinusoid whose frequency coincides exactly with DFT bin 3; energy confined to one conjugate pair.	{"N": 26, "bin_index": 3, "amplitude": 1.0}
S2	Non-bin-centered sinusoid	synthetic	26	4.27009e-18	0.70606	6.04777e-18	Sinusoid at a non-integer bin position	{"N": 26, "bin_offset": 3.5,

id	name	provenance	N	mean	std	r	description	params
							(3.5); worst-case leakage at the half-bin offset.	"amplitude": 1.0}
S3	Multi-component sinusoid with offset	synthetic	26	3.40588	0.851469	4	Three sinusoidal components plus a constant offset chosen so that mean/std = 4.0.	{"N": 26, "r": 4.0, "bins": [2, 5, 9], "amps": [1.0, 0.6, 0.3]}
S4	Broadband zero-mean sequence	synthetic	26	-2.98906e-17	1.10067	-2.71567e-17	White Gaussian sequence, mean removed. Spectrally flat in expectation; a lower bound on achievable compaction.	{"N": 26, "seed": 20260826}
S5	Damped transient	synthetic	26	1.09421e-17	0.246962	4.43068e-17	Exponentially damped oscillation; energy localised in time and therefore spread in frequency.	{"N": 26, "decay": 0.25, "bin_index": 4}

Table S4. Corpus statistics and DC energy share.

Source: *tables/table04_corpus_statistics.csv*

id	name	provenance	N	mean	std	r	energy	eta_dc	eps_parse val
S0	Recovered historical baseline	recovered	26	246.385	66.6391	3.6973	1.6938e+ 06	0.931834	2.74921e- 16
S1	Bin-centered sinusoid	synthetic	26	-8.54018e- 18	0.707107	-1.20776e- 17	13	1.48148e- 33	2.73286e- 16
S2	Non-bin-centered sinusoid	synthetic	26	4.27009e- 18	0.70606	6.04777e- 18	12.9615	1.48588e- 35	6.85242e- 16
S3	Multi-component sinusoid with offset	synthetic	26	3.40588	0.851469	4	320.45	0.941176	1.77386e- 16
S4	Broadband zero- mean sequence	synthetic	26	-2.98906e- 17	1.10067	-2.71567e- 17	31.4985	1.19841e- 33	1.01511e- 15
S5	Damped transient	synthetic	26	1.09421e- 17	0.246962	4.43068e- 17	1.58574	1.94324e- 33	2.80051e- 16

Table S5. Experiment A: measured against analytical DC energy share.

Source: tables/table05_experiment_a_dc_sweep.csv

r	mean	std	eta_dc_me asured	eta_dc_pr edicted	abs_error	rel_error	eta_total _7coef	eta_ac_7 coef	eps_l2_ful l_7coef	eps_l _7c
0	8.54018e- 18	0.851469	0	0	0	NaN	0.689655	0.689655	0.557086	0.55
0.5	0.425735	0.851469	0.2	0.2	1.94289e- 16	9.71445e- 16	0.751724	0.689655	0.498273	0.55
1	0.851469	0.851469	0.5	0.5	0	0	0.844828	0.689655	0.393919	0.55
2	1.70294	0.851469	0.8	0.8	1.11022e- 16	1.38778e- 16	0.937931	0.689655	0.249136	0.55
4	3.40588	0.851469	0.941176	0.941176	1.11022e- 16	1.17961e- 16	0.981744	0.689655	0.135113	0.55
8	6.81175	0.851469	0.984615	0.984615	0	0	0.995225	0.689655	0.069098	0.55

r	mean	std	eta_dc_me asured	eta_dc_pr edicted	abs_error	rel_error	eta_total _7coef	eta_ac_7 coef	eps_l2_ful l_7coef	eps_l _7c
16	13.6235	0.851469	0.996109	0.996109	1.11022e- 16	1.11456e- 16	0.998792	0.689655	0.0347501	0.55

Table S6. Coefficients required per energy threshold, both bases, all corpus members.

Source: tables/table06_energy_thresholds.csv

id	basis	threshold	achieved	n_coeff	coeff_ratio	eps_l2	rmse	snr_db
S0	total	0.5	0.931834	1	0.0384615	0.261087	66.6391	11.6643
S0	total	0.75	0.931834	1	0.0384615	0.261087	66.6391	11.6643
S0	total	0.8	0.931834	1	0.0384615	0.261087	66.6391	11.6643
S0	total	0.9	0.931834	1	0.0384615	0.261087	66.6391	11.6643
S0	total	0.95	0.964741	3	0.115385	0.187775	47.9271	14.5273
S0	total	0.99	0.990637	9	0.346154	0.0967602	24.6968	20.2861
S0	total	0.995	0.995081	14	0.538462	0.0701389	17.9021	23.0808
S0	total	0.999	0.999069	22	0.846154	0.0305122	7.78786	30.3105
S0	ac	0.5	0.696551	4	0.153846	0.550862	36.709	5.17914
S0	ac	0.75	0.806181	6	0.230769	0.440249	29.3378	7.12604
S0	ac	0.8	0.806181	6	0.230769	0.440249	29.3378	7.12604
S0	ac	0.9	0.911414	12	0.461538	0.297634	19.834	10.5264
S0	ac	0.95	0.960118	17	0.653846	0.199705	13.3082	13.9922
S0	ac	0.99	0.993648	23	0.884615	0.0796975	5.31097	21.9711
S0	ac	0.995	1	25	0.961538	9.47025e-15	6.31089e-13	280.473

S0	ac	0.999	1	25	0.961538	9.47025e-15	6.31089e-13	280.473
S1	total	0.5	1	2	0.0769231	3.00469e-15	2.12463e-15	290.444
S1	total	0.75	1	2	0.0769231	3.00469e-15	2.12463e-15	290.444
S1	total	0.8	1	2	0.0769231	3.00469e-15	2.12463e-15	290.444
S1	total	0.9	1	2	0.0769231	3.00469e-15	2.12463e-15	290.444
S1	total	0.95	1	2	0.0769231	3.00469e-15	2.12463e-15	290.444
S1	total	0.99	1	2	0.0769231	3.00469e-15	2.12463e-15	290.444
S1	total	0.995	1	2	0.0769231	3.00469e-15	2.12463e-15	290.444
S1	total	0.999	1	2	0.0769231	3.00469e-15	2.12463e-15	290.444
S1	ac	0.5	1	2	0.0769231	3.00847e-15	2.12731e-15	290.433
S1	ac	0.75	1	2	0.0769231	3.00847e-15	2.12731e-15	290.433
S1	ac	0.8	1	2	0.0769231	3.00847e-15	2.12731e-15	290.433
S1	ac	0.9	1	2	0.0769231	3.00847e-15	2.12731e-15	290.433
S1	ac	0.95	1	2	0.0769231	3.00847e-15	2.12731e-15	290.433
S1	ac	0.99	1	2	0.0769231	3.00847e-15	2.12731e-15	290.433
S1	ac	0.995	1	2	0.0769231	3.00847e-15	2.12731e-15	290.433
S1	ac	0.999	1	2	0.0769231	3.00847e-15	2.12731e-15	290.433
S2	total	0.5	0.814638	4	0.153846	0.430537	0.303985	7.31979
S2	total	0.75	0.814638	4	0.153846	0.430537	0.303985	7.31979
S2	total	0.8	0.814638	4	0.153846	0.430537	0.303985	7.31979
S2	total	0.9	0.906946	8	0.307692	0.305047	0.215382	10.3127
S2	total	0.95	0.959149	14	0.538462	0.202117	0.142707	13.8879

S2	total	0.99	0.990971	22	0.846154	0.095023	0.0670919	20.4434
S2	total	0.995	0.997033	24	0.923077	0.0544735	0.0384615	25.2763
S2	total	0.999	1	25	0.961538	2.95471e-15	2.08621e-15	290.59

Showing the first 40 of 96 rows; the complete table is in `tables/table06_energy_thresholds.csv`.

Table S7. Retention curves for all strategy and basis combinations.

Source: `tables/table07_selection_curves.csv`

n_co eff	coeff_rati o	eta_total	eta_ac	eps_l2	rmse	snr_db	max_ imag	strategy	basis	id
1	0.0384615	0.931834	0	0.261087	66.6391	11.6643	0	lowpass	total	S0
3	0.115385	0.935683	0.0564705	0.253608	64.7302	11.9167	0	lowpass	total	S0
5	0.192308	0.96859	0.539216	0.177228	45.2353	15.0293	0	lowpass	total	S0
7	0.269231	0.976063	0.648847	0.154715	39.4892	16.2093	0	lowpass	total	S0
9	0.346154	0.990637	0.862652	0.0967602	24.6968	20.2861	0	lowpass	total	S0
11	0.423077	0.991747	0.87893	0.0908455	23.1872	20.8339	0	lowpass	total	S0
13	0.5	0.993773	0.908652	0.0789105	20.1409	22.0573	0	lowpass	total	S0
15	0.576923	0.994864	0.92466	0.0716634	18.2912	22.894	0	lowpass	total	S0
17	0.653846	0.995362	0.931966	0.0681001	17.3817	23.337	0	lowpass	total	S0
19	0.730769	0.996398	0.947151	0.0600208	15.3195	24.434	0	lowpass	total	S0
21	0.807692	0.997695	0.966192	0.0480059	12.2529	26.3741	0	lowpass	total	S0
23	0.884615	0.998128	0.972544	0.043262	11.0421	27.2779	0	lowpass	total	S0
25	0.961538	0.998881	0.983583	0.033453	8.53846	29.5113	0	lowpass	total	S0
26	1	1	1	2.50703e-15	6.39889e-13	292.017	0	lowpass	total	S0

n_co eff	coeff_ratio	eta_total	eta_ac	eps_l2	rmse	snr_db	max_imag	strategy	basis	id
2	0.0769231	0.00384939	0.0564705	0.971354	64.7302	0.252445	0	lowpass	ac	S0
4	0.153846	0.0367564	0.539216	0.67881	45.2353	3.36503	0	lowpass	ac	S0
6	0.230769	0.0442295	0.648847	0.592582	39.4892	4.54503	0	lowpass	ac	S0
8	0.307692	0.0588038	0.862652	0.370605	24.6968	8.62177	0	lowpass	ac	S0
10	0.384615	0.0599134	0.87893	0.347951	23.1872	9.16963	0	lowpass	ac	S0
12	0.461538	0.0619395	0.908652	0.302238	20.1409	10.393	0	lowpass	ac	S0
14	0.538462	0.0630307	0.92466	0.274481	18.2912	11.2297	0	lowpass	ac	S0
16	0.615385	0.0635287	0.931966	0.260833	17.3817	11.6727	0	lowpass	ac	S0
18	0.692308	0.0645639	0.947151	0.229888	15.3195	12.7697	0	lowpass	ac	S0
20	0.769231	0.0658618	0.966192	0.18387	12.2529	14.7098	0	lowpass	ac	S0
22	0.846154	0.0662947	0.972544	0.1657	11.0421	15.6136	0	lowpass	ac	S0
24	0.923077	0.0670472	0.983583	0.12813	8.53846	17.847	0	lowpass	ac	S0
25	0.961538	0.0681663	1	9.47025e-15	6.31089e-13	280.473	0	lowpass	ac	S0
1	0.0384615	0.931834	0	0.261087	66.6391	11.6643	0	magnitude	total	S0
3	0.115385	0.964741	0.482746	0.187775	47.9271	14.5273	0	magnitude	total	S0
5	0.192308	0.979315	0.696551	0.143823	36.709	16.8434	0	magnitude	total	S0
7	0.269231	0.986788	0.806181	0.114943	29.3378	18.7903	0	magnitude	total	S0
9	0.346154	0.990637	0.862652	0.0967602	24.6968	20.2861	0	magnitude	total	S0
11	0.423077	0.992664	0.892374	0.0856533	21.8619	21.3451	0	magnitude	total	S0
13	0.5	0.993961	0.911414	0.0777082	19.834	22.1907	0	magnitude	total	S0

n_coeff	coeff_ratio	eta_total	eta_ac	eps_l2	rmse	snr_db	max_imag	strategy	basis	id
14	0.538462	0.995081	0.927831	0.0701389	17.9021	23.0808	0	magnitude	total	S0
16	0.615385	0.99619	0.94411	0.0617239	15.7543	24.1909	0	magnitude	total	S0
18	0.692308	0.997281	0.960118	0.0521404	13.3082	25.6565	0	magnitude	total	S0
20	0.769231	0.998317	0.975303	0.0410304	10.4725	27.7379	0	magnitude	total	S0
22	0.846154	0.999069	0.986342	0.0305122	7.78786	30.3105	0	magnitude	total	S0
24	0.923077	0.999567	0.993648	0.020808	5.31097	33.6354	0	magnitude	total	S0

Showing the first 40 of 324 rows; the complete table is in `tables/table07_selection_curves.csv`.

Table S8. DFT against DCT-II coefficient counts at fixed thresholds.

Source: `tables/table08_dft_vs_dct.csv`

id	transform	threshold	n_coeff	eps_l2	mean_removed
S0	DCT-II	0.9	10	0.299484	True
S0	DFT	0.9	12	0.297634	True
S0	DCT-II	0.95	14	0.199443	True
S0	DFT	0.95	17	0.199705	True
S0	DCT-II	0.99	18	0.0951672	True
S0	DFT	0.99	23	0.0796975	True
S1	DCT-II	0.9	2	0.253457	True
S1	DFT	0.9	2	3.00847e-15	True
S1	DCT-II	0.95	3	0.146825	True
S1	DFT	0.95	2	3.00847e-15	True
S1	DCT-II	0.99	5	0.0699666	True

id	transform	threshold	n_coeff	eps_l2	mean_removed
S1	DFT	0.99	2	3.00847e-15	True
S2	DCT-II	0.9	2	0.291912	True
S2	DFT	0.9	8	0.305047	True
S2	DCT-II	0.95	3	0.163628	True
S2	DFT	0.95	14	0.202117	True
S2	DCT-II	0.99	5	0.0832141	True
S2	DFT	0.99	22	0.095023	True
S3	DCT-II	0.9	3	0.274561	True
S3	DFT	0.9	4	0.249136	True
S3	DCT-II	0.95	4	0.209276	True
S3	DFT	0.95	6	2.98348e-15	True
S3	DCT-II	0.99	9	0.0778313	True
S3	DFT	0.99	6	2.98348e-15	True
S4	DCT-II	0.9	12	0.310769	True
S4	DFT	0.9	18	0.267445	True
S4	DCT-II	0.95	15	0.211626	True
S4	DFT	0.95	20	0.205312	True
S4	DCT-II	0.99	20	0.0892631	True
S4	DFT	0.99	23	0.0518493	True
S5	DCT-II	0.9	7	0.302563	True

id	transform	threshold	n_coeff	eps_l2	mean_removed
S5	DFT	0.9	16	0.29582	True
S5	DCT-II	0.95	9	0.16491	True
S5	DFT	0.95	20	0.206343	True
S5	DCT-II	0.99	10	0.0747719	True
S5	DFT	0.99	24	0.090032	True

Table S9. Spectral leakage and window characteristics.

Source: tables/table09_leakage_and_windows.csv

leakage_	spectral_flat	bin_o	bin_cent	eta_dc	experiment	mainlobe_	peak_side	window	cohe
energy_f	ness	ffset	ered			halfwidth_	lobe_db		ga
raction						bins			
0	9.89619e-28	3	True	1.48148e-33	leakage	NaN	NaN	NaN	
0.042222	0.00570632	3.125	False	9.14785e-34	leakage	NaN	NaN	NaN	
0.162085	0.0651014	3.25	False	9.20677e-34	leakage	NaN	NaN	NaN	
0.549223	0.280729	3.5	False	1.48588e-35	leakage	NaN	NaN	NaN	
0.549223	0.280729	3.5	NaN	NaN	windows	1.00195	-13.2182	rectangular	
0.518356	0.000153407	3.5	NaN	NaN	windows	2.00195	-31.4706	hann	
0.505142	0.0111474	3.5	NaN	NaN	windows	2.00195	-41.4056	hamming	
0.549911	4.58337e-05	3.5	NaN	NaN	windows	3.00195	-58.1588	blackman	

Table S10. Noise robustness summary, 100 realizations per level.

Source: tables/table10_noise_summary.csv

snr_db	n	confidence	eta_dc_mean	eta_dc_sd	eta_dc_ci_lo	eta_dc_ci_hi	eta_total_mean	eta_total_sd	eta_total_ci_lo	eta_total_ci_hi
30	100	0.95	0.931656	0.000664379	0.931526	0.931786	0.991713	0.000290383	0.991656	0.991770
20	100	0.95	0.931076	0.00264132	0.930558	0.931593	0.991356	0.00103492	0.991153	0.991559
10	100	0.95	0.926527	0.00739464	0.925078	0.927976	0.98812	0.00286442	0.987558	0.988692
0	100	0.95	0.875735	0.0264443	0.870552	0.880918	0.958114	0.0160284	0.954973	0.961255

Table S11. Parseval discrepancy against transform length and precision.

Source: tables/table11_numerical_precision.csv

N	precision	eps_parseval
16	float32	6.0617e-08
16	float64	1.77081e-16
26	float32	1.76441e-08
26	float64	1.37461e-16
64	float32	4.17844e-08
64	float64	0
256	float32	5.45807e-08
256	float64	0
1024	float32	6.07031e-08
1024	float64	0
4096	float32	5.11822e-08

N **precision** **eps_parseval**

4096 float64 0

Table S12. All numerical validation gates, measured values and tolerances.

Source: tables/table12_validation_checks.csv

check	measured	tolerance	passed	note
direct DFT vs FFT (max abs diff)	1.91181e-12	1e-09	True	NaN
direct DFT vs FFT (max rel diff)	1.52175e-15	1e-12	True	NaN
IDFT(DFT(x)) round-trip	1.69905e-12	1e-09	True	NaN
Hermitian symmetry violation	1.813e-12	1e-09	True	NaN
Nyquist bin imaginary part	6.54789e-13	1e-09	True	N even => X[N/2] real
DC bin imaginary part	0	1e-09	True	NaN
legacy transform == mirrored unitary DFT	1.813e-12	1e-09	True	confirms the 2013/14 spectrum is frequency- reversed
legacy == unitary (magnitudes unaffected)	0	1e-09	True	NaN
Parseval discrepancy, float64	2.74921e-16	1e-12	True	NaN
eta_DC measured vs $r^2/(1+r^2)$	3.33067e-16	1e-12	True	NaN
DCT-II Parseval	1.37461e-16	1e-12	True	NaN
DCT-II round-trip	9.09495e-13	1e-09	True	NaN
11-coefficient reconstruction is real	7.01661e-13	1e-09	True	NaN
full-mask reconstruction == x	1.67688e-12	1e-09	True	NaN
historical: E (sample domain)	0.000118064	0.001	True	NaN

check	measured	tolerance	passed	note
historical: H (transform domain)	0.000118064	0.001	True	NaN
historical: E1 (k=5, 11 coefficients)	0.000106414	0.001	True	NaN
historical: 100*E1/E	2.93187e-06	0.001	True	NaN
historical: E2 (all coefficients)	0.000118064	0.001	True	NaN
historical: 100*E2/E	1.42109e-16	0.001	True	NaN
historical: 0.95E	6.83654e-05	0.001	True	NaN
historical: yc[k], ys[k] for k = 0..14 (max abs deviation)	0.319962	0.5	True	yc[0] deviates by 0.320 because it is printed to 4 s.f.
historical: printed samples x[0..14] (max abs deviation)	0	0.5	True	NaN
identity: eps_L2 == sqrt(1 - eta)	2.50703e-15	1e-12	True	NaN
identity: 1-eta_total == (1-eta_AC)(1-eta_DC)	3.33067e-16	1e-12	True	NaN
identity: eps_full == eps_AC / sqrt(1+r^2)	8.32667e-17	1e-12	True	NaN
magnitude ranking attains the true optimum at every budget	2.22045e-16	1e-12	True	exhaustive over 16384 conjugate-symmetric sets
window sidelobe vs literature: rectangular	0.0456563	1.5	True	measured -13.3 dB, literature - 13.3 dB
window sidelobe vs literature: hann	0.0325882	1.5	True	measured -31.5 dB, literature - 31.5 dB
window sidelobe vs literature: hamming	0.749281	1.5	True	measured -42.4 dB, literature - 41.7 dB
window sidelobe vs literature:	0.0101484	1.5	True	measured -58.1 dB, literature -

check	measured	tolerance	passed	note
blackman				58.1 dB
cumulative-energy mask reaches its threshold	0	1e-12	True	NaN
cumulative mask is conjugate- symmetric	4.73399e-13	1e-09	True	NaN